# Taller Generativos

*Valero Laparra*

# Can generative models improve ML performance?

# PROBLEM

**In general machine learning is…**

- **Data** $$[\{y_1, X_1\}, \{y_2, X_2\}, \ldots \{y_N, X_N\}] = [\mathbf{y}, \mathbf{X}]$$

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- **Model** $$\hat{y}_i = f(X_i, \theta)$$

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**- Learning** $$\min_{\theta} D(\hat{\mathbf{y}}, \mathbf{y})$$

We need **enough** samples (N) to make the model to generalize to new samples (i)

# Question

**Can we generate new samples artificially?**

**(and improve the model)**

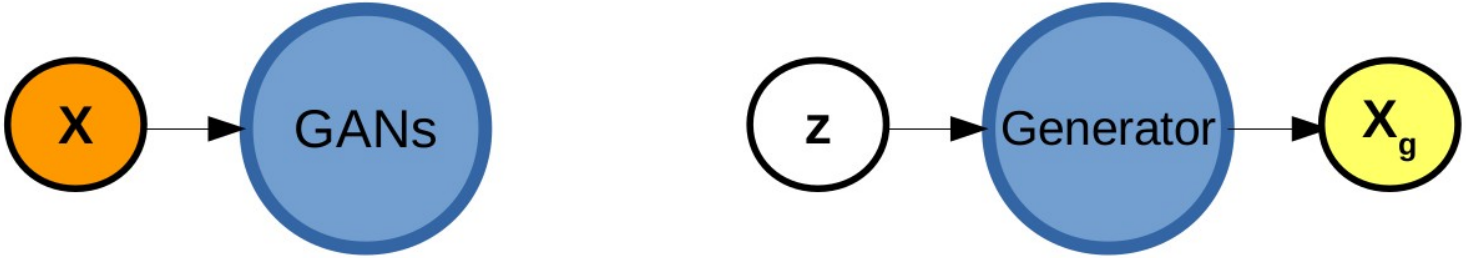
$$[\mathbf{y}_g, \mathbf{X}_g]$$

$$[\mathbf{y}_T, \mathbf{X}_T] = [\mathbf{y}, \mathbf{X}, \mathbf{y}_g, \mathbf{X}_g]$$

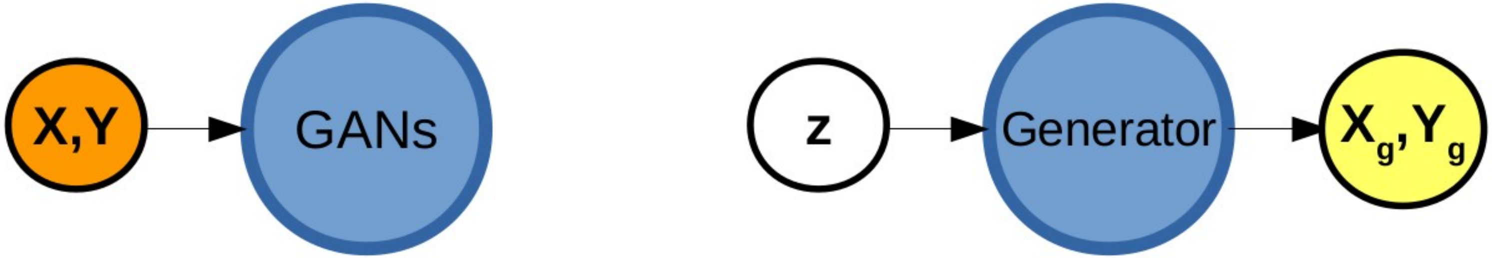
# GANs: options

## OPTIONS to apply GANs

**OPT1: GANs** for generating inputs



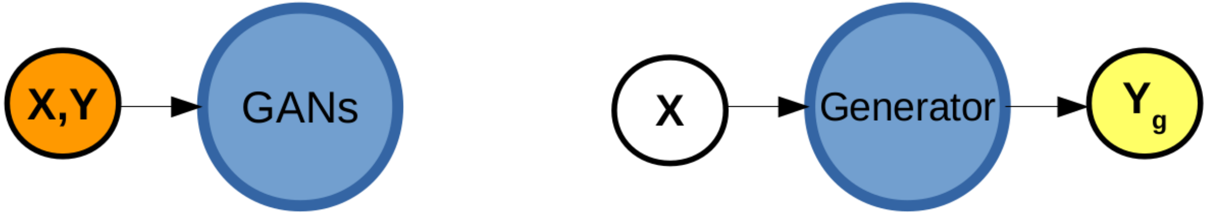

**OPT2: GANs** for generating input-output pairs



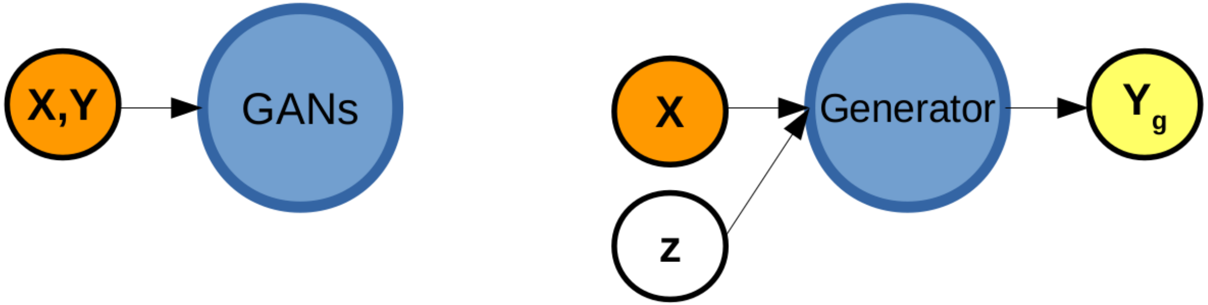
# GANs: options

## OPTIONS to apply Conditional GANs

**OPT3: Conditional GANs** for generating input-output pairs, controlling the input features




**OPT4: Conditional GANs** for generating input-output pairs, controlling the input features. Stocastic component

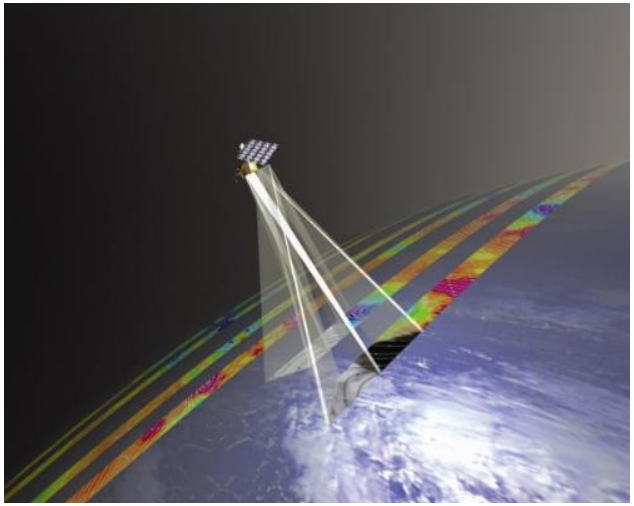


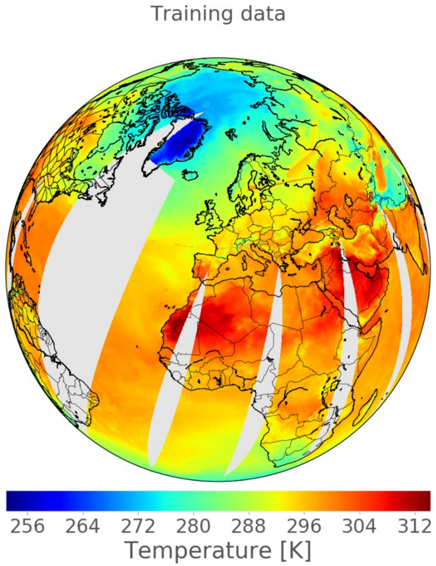
# GANs: options

**Real problem:**
Estimate land surface temperature (LST) from IASI observations

$\mathbf{x}$

$\mathbf{y}$





# REAL PROBLEM

- **Data: { y=LST, X=IASI }**

$$[\{y_1, X_1\}, \{y_2, X_2\}, \ldots \{y_N, X_N\}] = [\mathbf{y}, \mathbf{X}]$$

- **Model: Artificial Neural Network (3 layers)**

$$\hat{y}_i = f(X_i, \theta)$$

- **Learning: minimize MAE**

$$\min_{\theta} D(\hat{\mathbf{y}}, \mathbf{y})$$

# REAL PROBLEM

- **Data: { y=LST, X=IASI }**

$$[\{y_1, X_1\}, \{y_2, X_2\}, \dots \{y_N, X_N\}] = [\mathbf{y}, \mathbf{X}]$$

$$[\{y_{g1}, X_{g1}\}, \{y_{g2}, X_{g2}\}, \dots \{y_{gN}, X_{gN}\}] = [\mathbf{y}_g, \mathbf{X}_g]$$

- **Model: Artificial Neural Network (3 layers)**

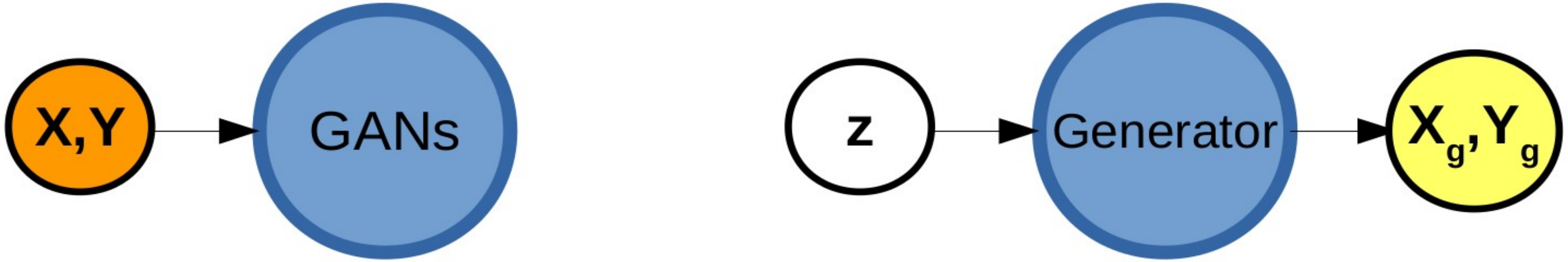
$$\hat{y}_i = f(X_i, \theta)$$

- **Learning: minimize MAE**

$$\min_\theta D(\hat{\mathbf{y}}, \mathbf{y})$$

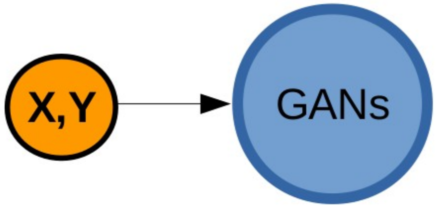
# GANs: options

**OPT2: GANs** for generating input-output pairs



# GANs: procedure

**STEP 1:** Train GANs



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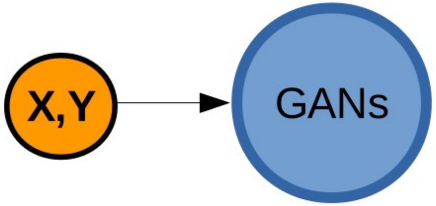
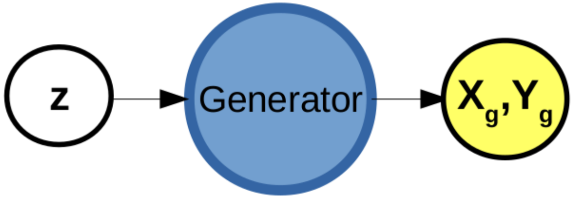



**STEP 2:** Synthetic data generation



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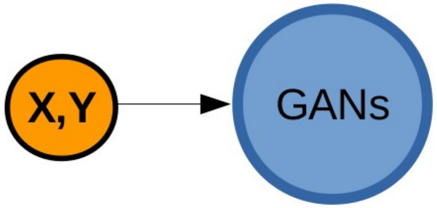
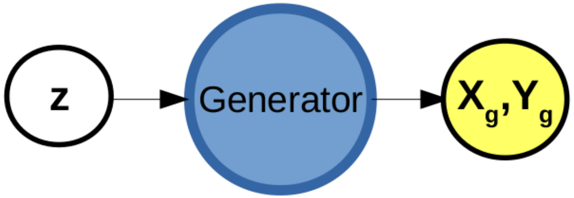
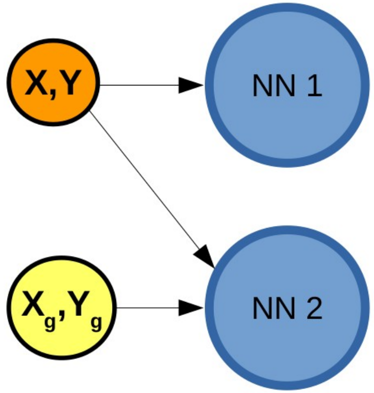



**STEP 2:** Synthetic data generation




**STEP 3:** NN training



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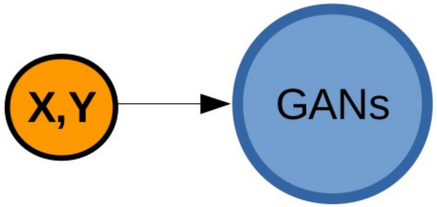


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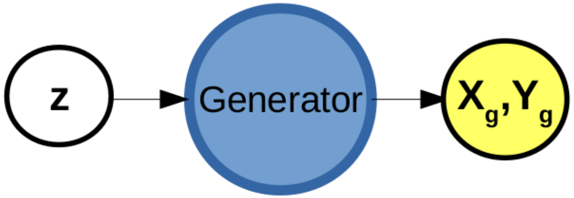


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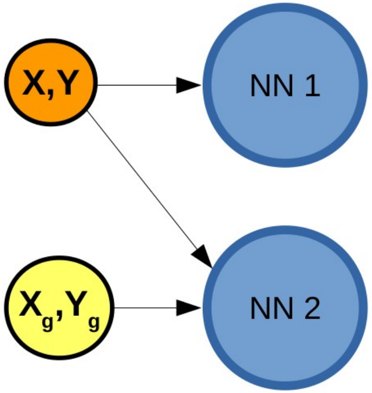
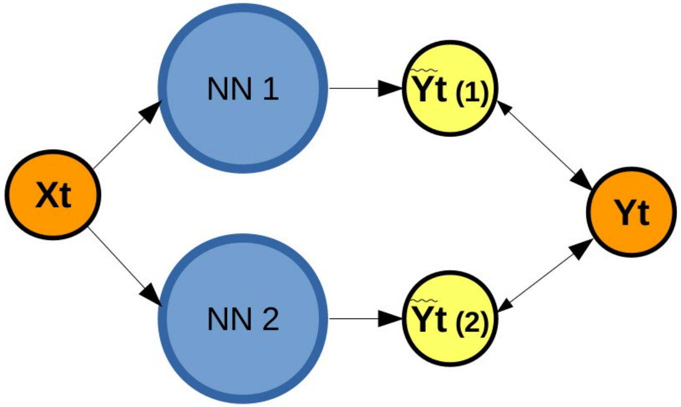



**STEP 4:** NN evaluation



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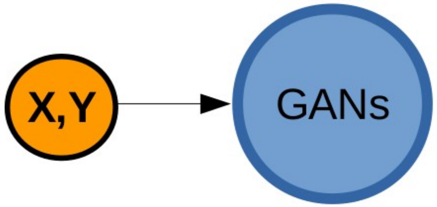
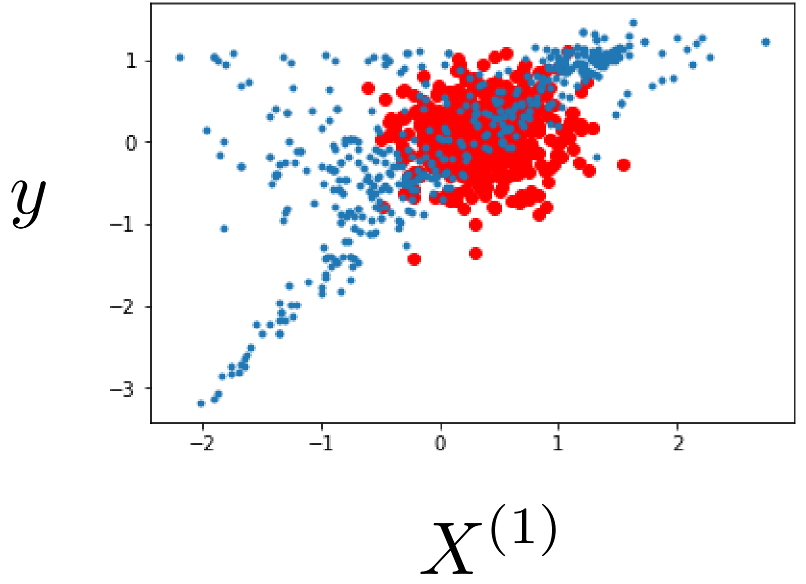



**Step 1**

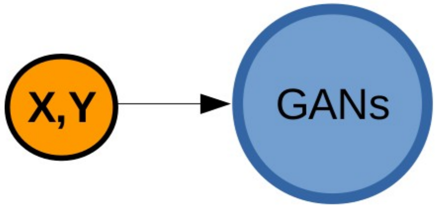



Reales {y,X}
Generados {y_g,X_g}

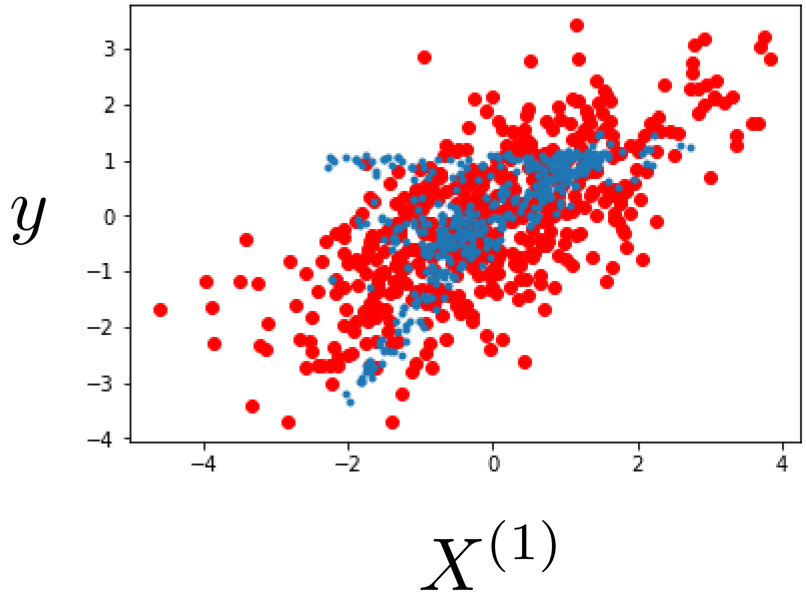
```
0 [D samples: 250, G samples: 500]
[D loss: 0.748248, acc.: 42.00%] [G loss: 0.730932, G acc: 45.60%]
[D2 loss: 0.687066, acc2.: 60.10%] [G2 loss: 0.702408, G2 acc: 49.00%]
```

# GANs: procedure

**STEP 1:** Train GANs




**Step 100**

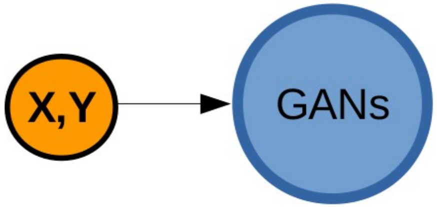



Reales {y,X}
Generados {y_g,X_g}

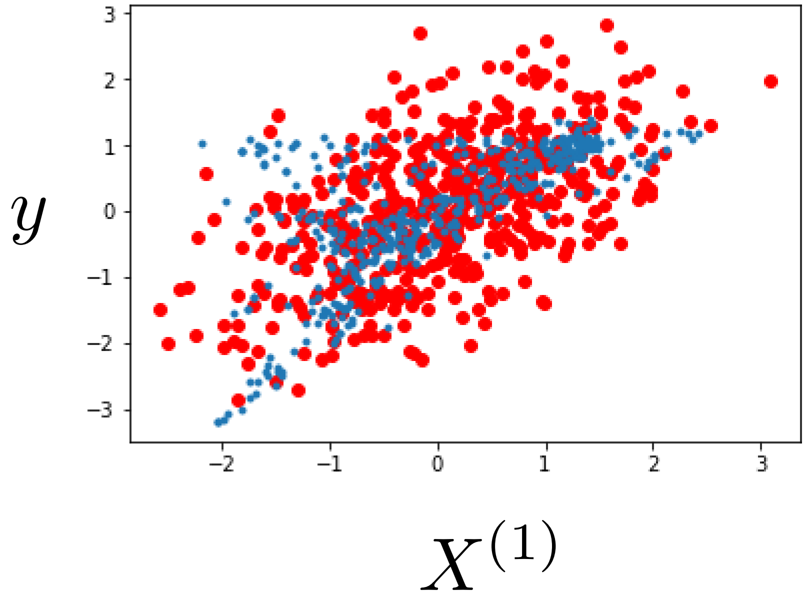
```
100 [D samples: 111, G samples: 439]
[D loss: 0.695746, acc.: 45.95%] [G loss: 0.771487, G acc: 13.44%]
[D2 loss: 0.678436, acc2.: 56.60%] [G2 loss: 0.760929, G2 acc: 16.80%]
```

# GANs: procedure

**STEP 1:** Train GANs




**Step 300**

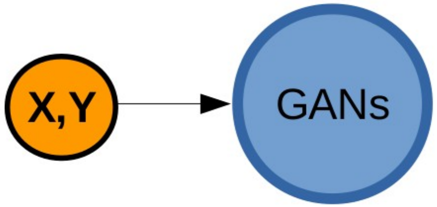



Reales {y,X}
Generados {y_g,X_g}

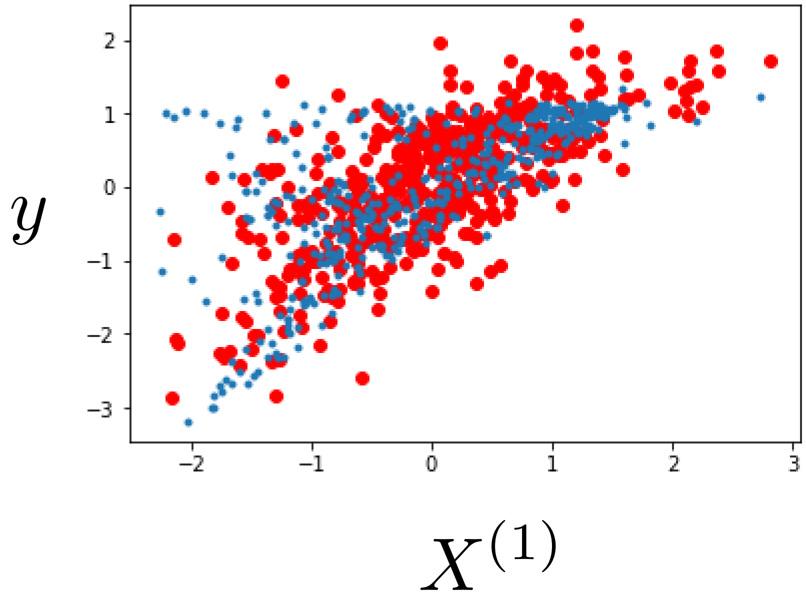
```
300 [D samples: 69, G samples: 261]
[D loss: 0.664405, acc.: 65.22%] [G loss: 0.701818, G acc: 46.74%]
[D2 loss: 0.655907, acc2.: 69.20%] [G2 loss: 0.698733, G2 acc: 47.20%]
```

# GANs: procedure

**STEP 1:** Train GANs




**Step 900**

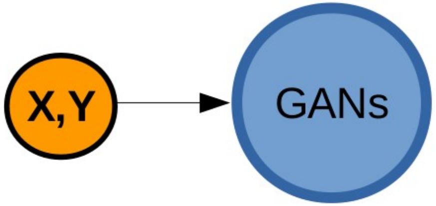



**Reales {y,X}**
**Generados {y_g,X_g}**

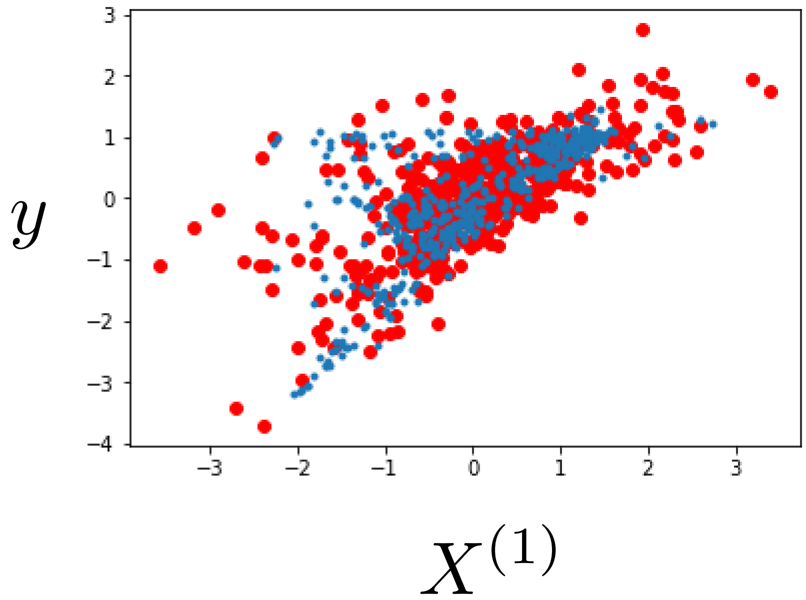
```
900 [D samples: 53, G samples: 453]
[D loss: 0.608968, acc.: 76.42%] [G loss: 0.821709, G acc: 8.83%]
[D2 loss: 0.596355, acc2.: 80.30%] [G2 loss: 0.806010, G2 acc: 11.60%]
```

# GANs: procedure

**STEP 1:** Train GANs




**Step 1300**

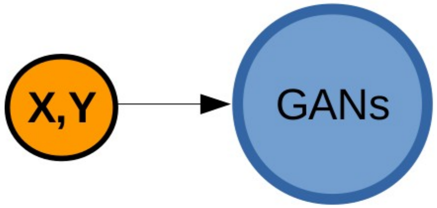



**Reales {y,X}**
**Generados {y_g,X_g}**

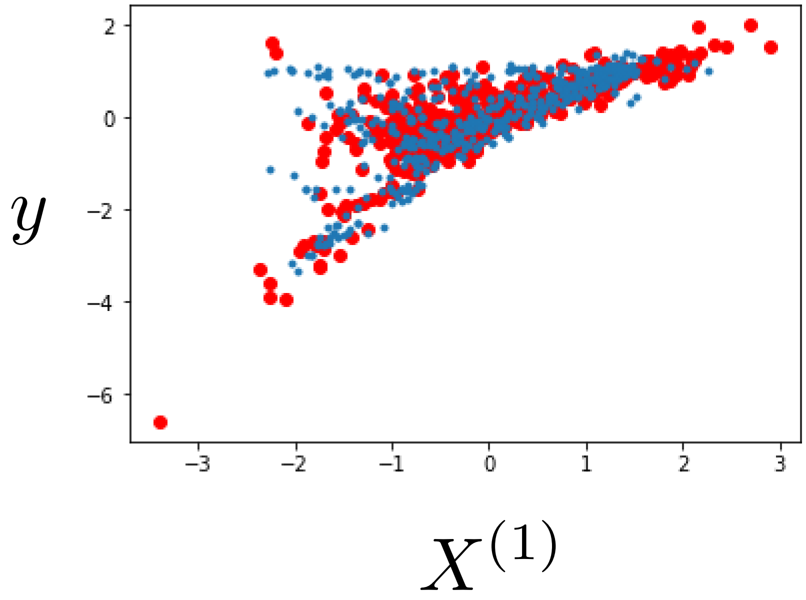
```
1300 [D samples: 60, G samples: 438]
[D loss: 0.586450, acc.: 74.17%] [G loss: 0.851556, G acc: 8.90%]
[D2 loss: 0.568890, acc2.: 79.10%] [G2 loss: 0.843278, G2 acc: 9.80%]
```

# GANs: procedure

**STEP 1:** Train GANs




**Step 9100**

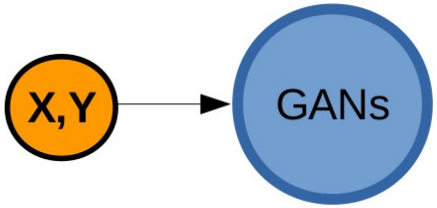



Reales {y,X}
Generados {y_g,X_g}

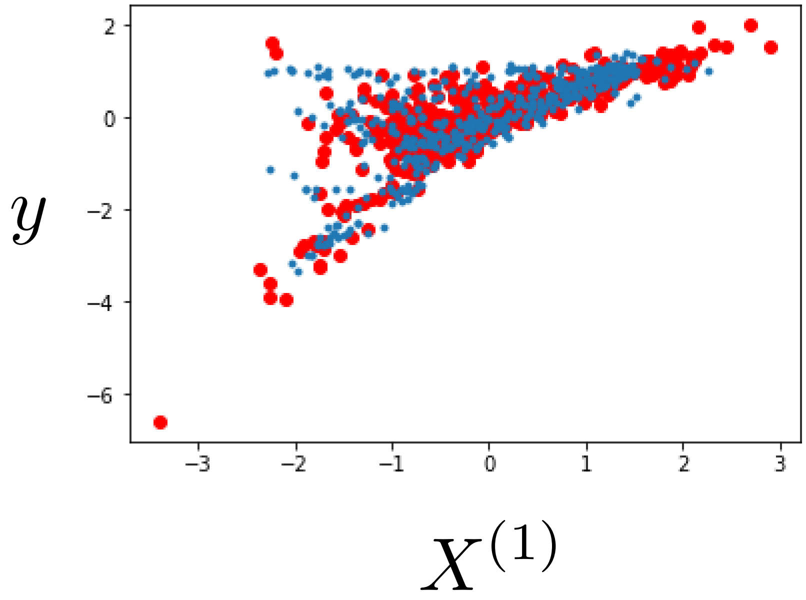
```
9100 [D samples: 28, G samples: 429]
[D loss: 0.407406, acc.: 80.36%] [G loss: 1.365164, G acc: 13.05%]
[D2 loss: 0.381066, acc2.: 88.80%] [G2 loss: 1.289948, G2 acc: 14.00%]
```

# GANs: procedure

**STEP 1:** Train GANs




**Step 9100**

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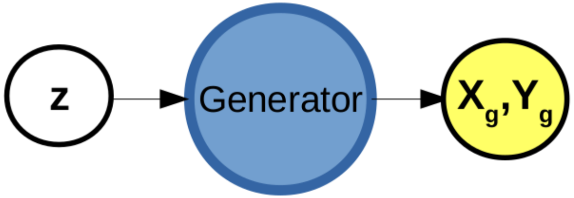
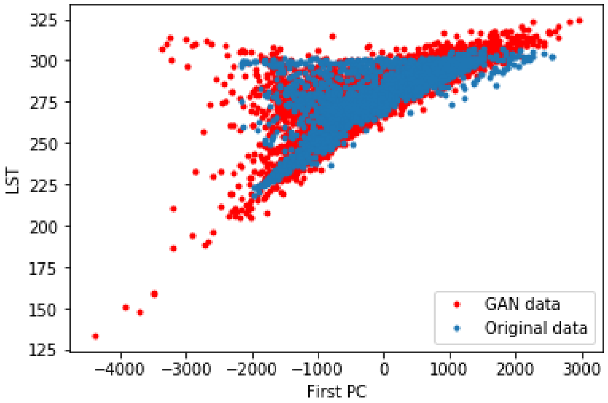
# GANs: procedure

**STEP 2:** Synthetic data generation

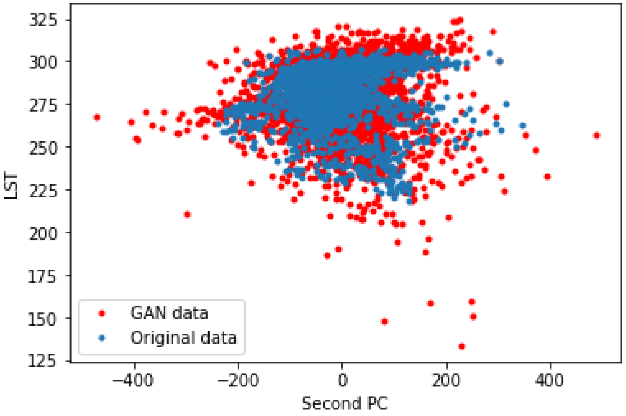



**Reales {y,X}**

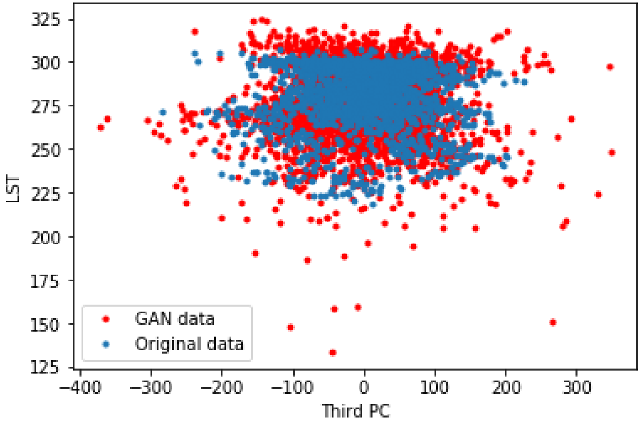
**Generados {y_g,X_g}**




$X^{(1)}$

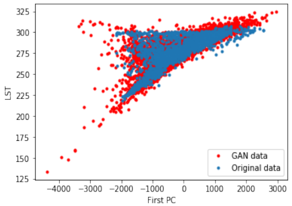



$X^{(2)}$

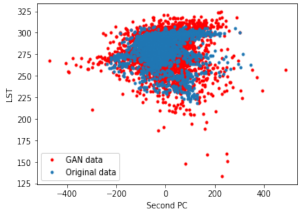



$X^{(3)}$

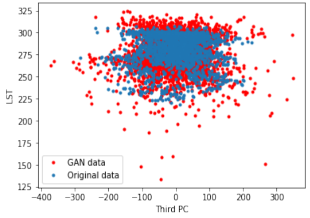
# GANs: procedure




$X^{(1)}$




$X^{(2)}$

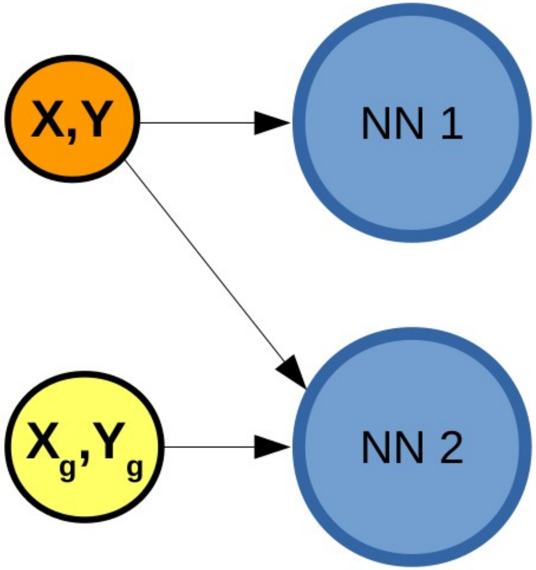



$X^{(3)}$

**STEP 3:** NN training

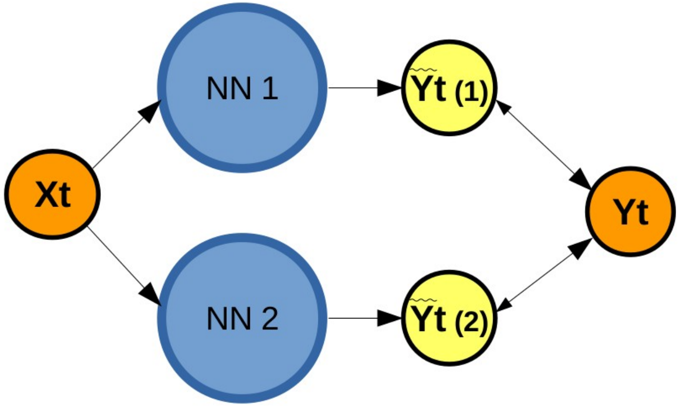
Reales {y,X}

Generados {y_g,X_g}



# GANs: procedure

**STEP 4:** NN evaluation



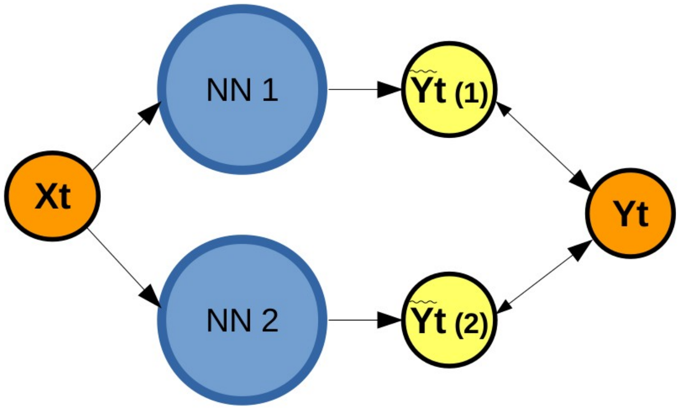

**Errors**

Only real data = 4.81K

Real + GANs data = 3.46 K

# GANs: procedure

**STEP 4:** NN evaluation




**Errors**

Only real data  = 4.81K

Real + GANs data =  3.46 K

Real + RBIG data =  3.62 K

**RBIG**

# CONCLUSIONS (ANSWER)

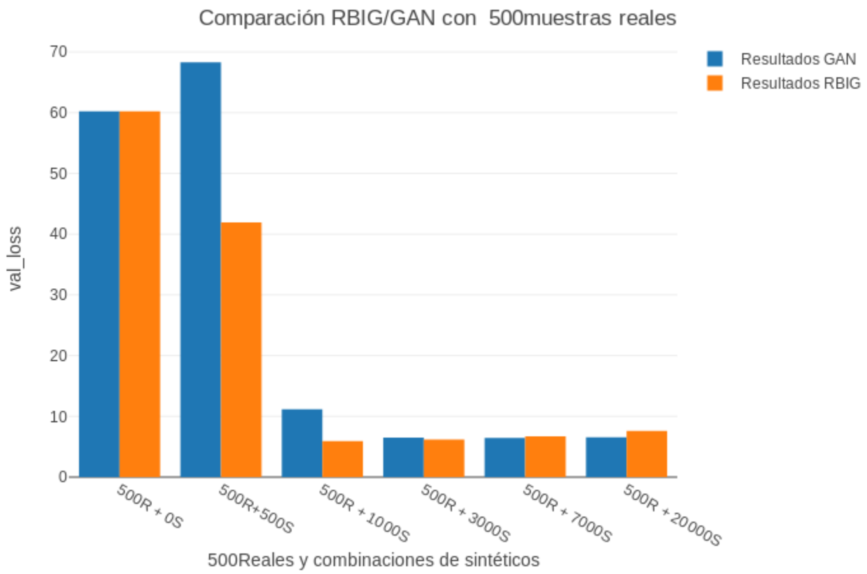
**Can we generate new samples artificially?**

**(and improve the model)**

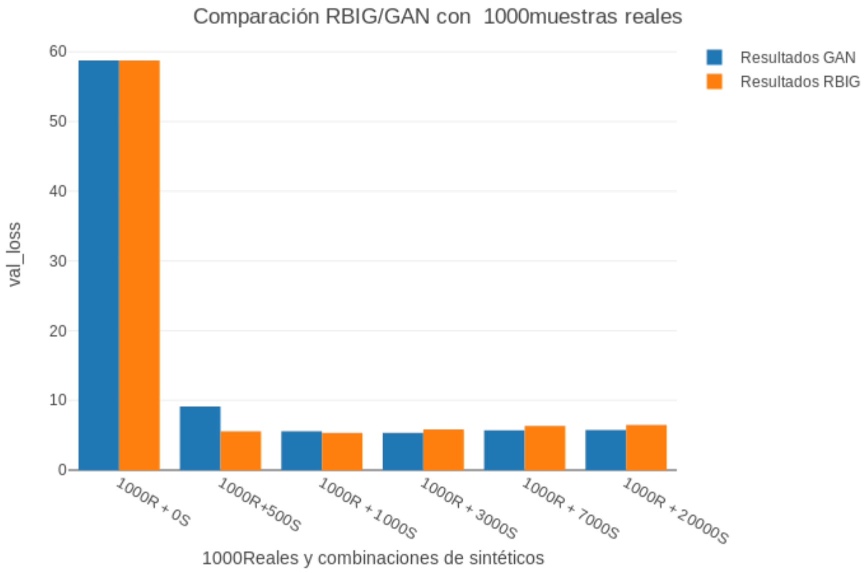
**yes**

# GANs vs RBIG

## 500 datos reales

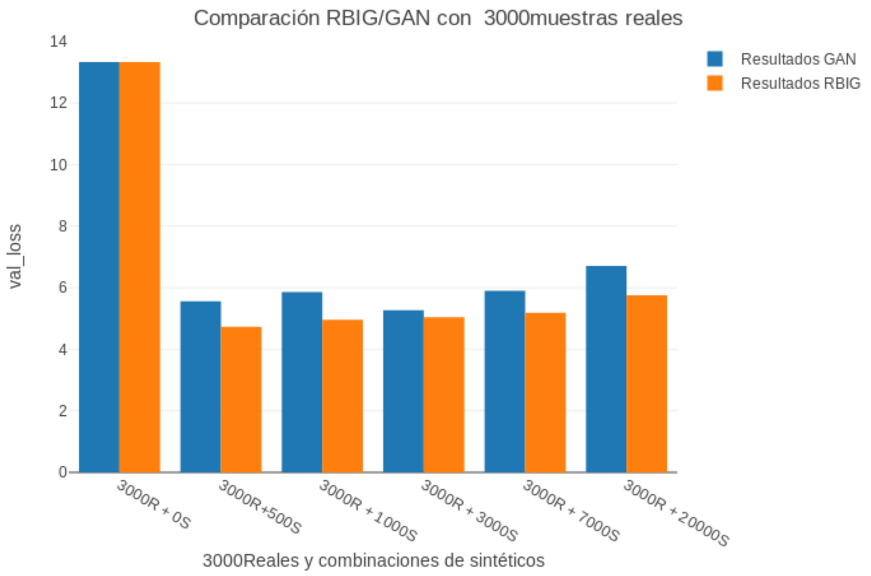



## 1000 datos reales

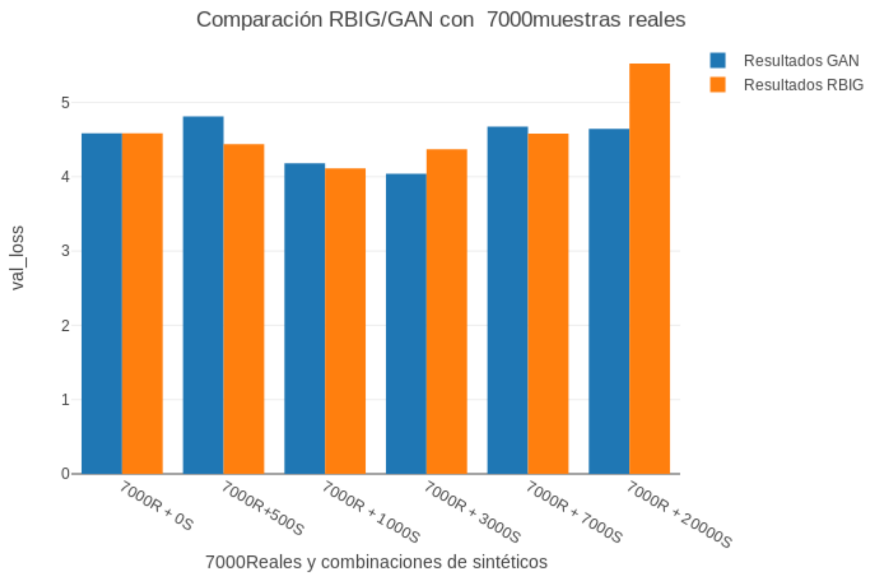


# GANs vs RBIG

## 3000 datos reales

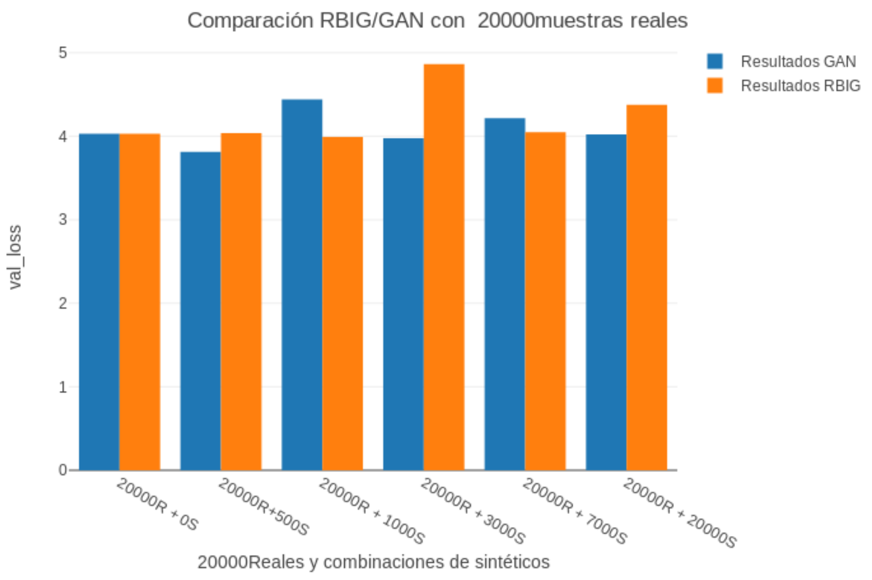



## 7000 datos reales



# GANs vs RBIG

## 20000 datos reales




## TODOS